

Low-Rank Tensor Approximation & Image Inpainting

Team 4 TP406 2026

1. Introduction & Motivation

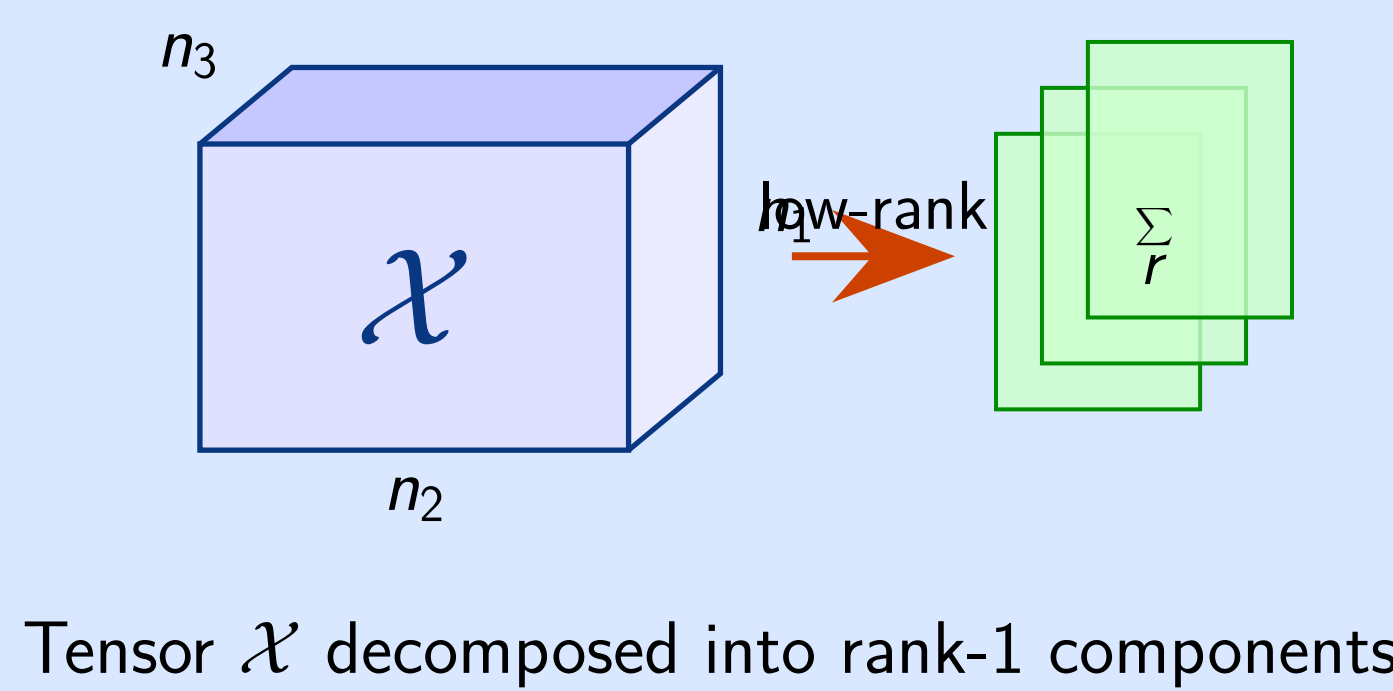
- ▶ Real-world data naturally arise as **multi-way arrays (tensors)**: RGB images, videos, hyperspectral data
- ▶ Low-rank decomposition yields **compact, structured representations**

Project Objective: Given $\mathcal{T} \in \mathbb{C}^{n_1} \otimes \mathbb{C}^{n_2} \otimes \mathbb{C}^{n_3}$,

$$\hat{\mathcal{T}} \min_{\text{low-rank}} \|\mathcal{T} - \hat{\mathcal{T}}\|_F$$

Contributions:

- ▶ CP-ALS decomposition from scratch
- ▶ Tucker decomposition (HOSVD / HOOI)
- ▶ Mathematical validation & convergence analysis
- ▶ Application to image inpainting



Tensor $\mathcal{X}$ decomposed into rank-1 components

2. CP Decomposition (Canonical Polyadic / ALS)

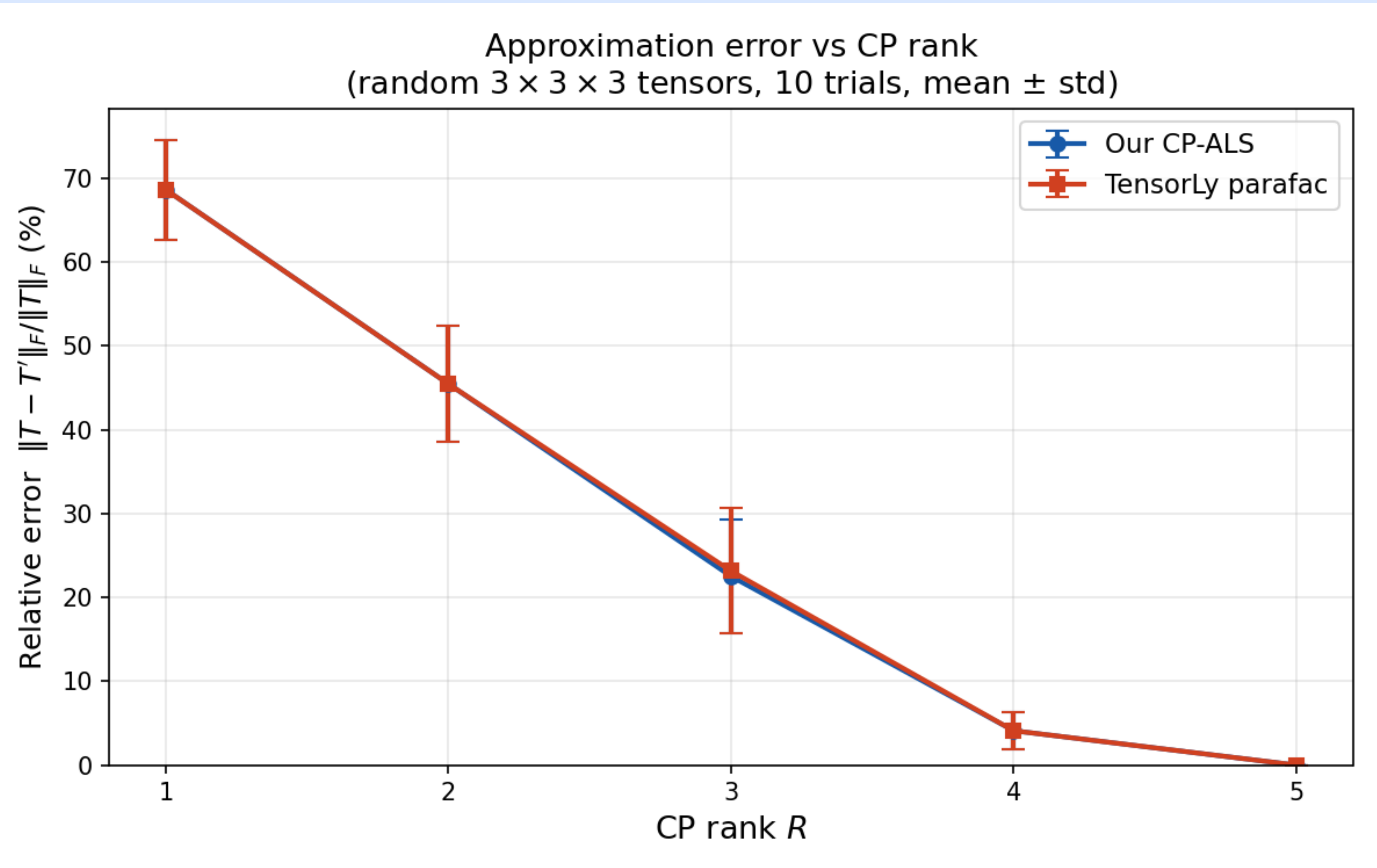
Approximate as a sum of R rank-one tensors:

$$\mathcal{X} \approx \sum_{r=1}^R \mathbf{a}_r \otimes \mathbf{b}_r \otimes \mathbf{c}_r$$

ALS mode-1 update:

$$A \leftarrow X_{(1)} (C \odot B) (B^T B * C^T C)^\dagger$$

- ▶ Fix $B, C \rightarrow$ update A ; cycle all modes
- ▶ Compact, interpretable representation
- ▶ Tensor rank R controls compression



CP-ALS matches TensorLy: relative error vs. rank R

3. Tucker Decomposition (HOSVD / HOOI)

Core tensor $\mathcal{G}$ contracted with factor matrices:

$$\mathcal{X} \approx \mathcal{G} \times_1 A \times_2 B \times_3 C$$

HOSVD init & HOOI refinement:

$$A^{(n)} = \text{trunc-SVD}_{r_n}(X_{(n)})$$
$$\mathcal{G} = \mathcal{X} \times_1 A^T \times_2 B^T \times_3 C^T$$

- ▶ Multi-linear rank (r_1, r_2, r_3) — more flexible than CP
- ▶ HOSVD gives closed-form init; HOOI iterates to optimum

figures/tucker_rank_sweep.png

Tucker (Lena): PSNR & compression ratio vs. rank

Mathematical Analysis, Inpainting & Results

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4. Mathematical Analysis & Validation

Optimization objective:

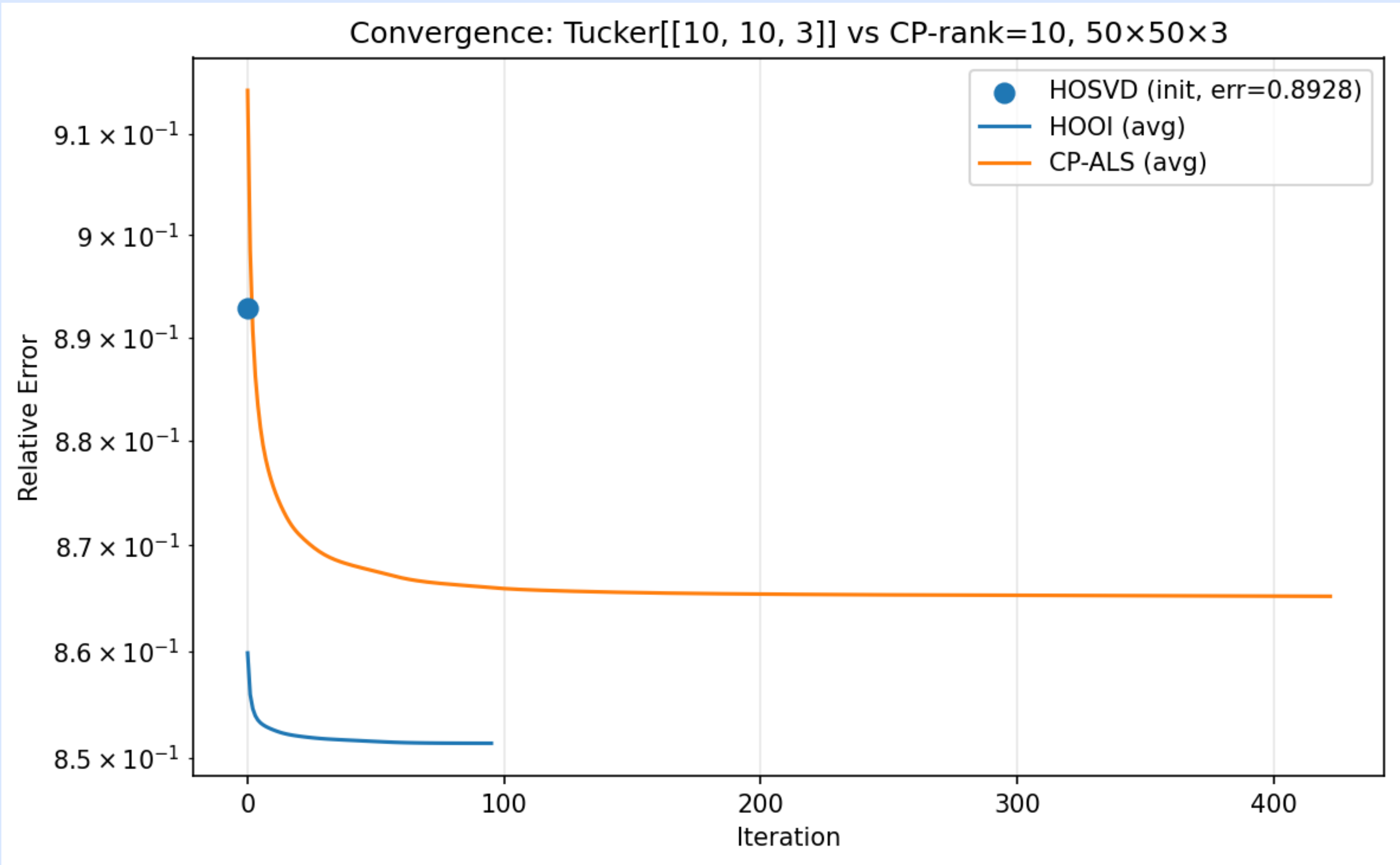
$$\min_{A,B,C} \left\| \mathcal{X} - \sum_{r=1}^R \mathbf{a}_r \otimes \mathbf{b}_r \otimes \mathbf{c}_r \right\|_F^2$$

Evaluation metrics:

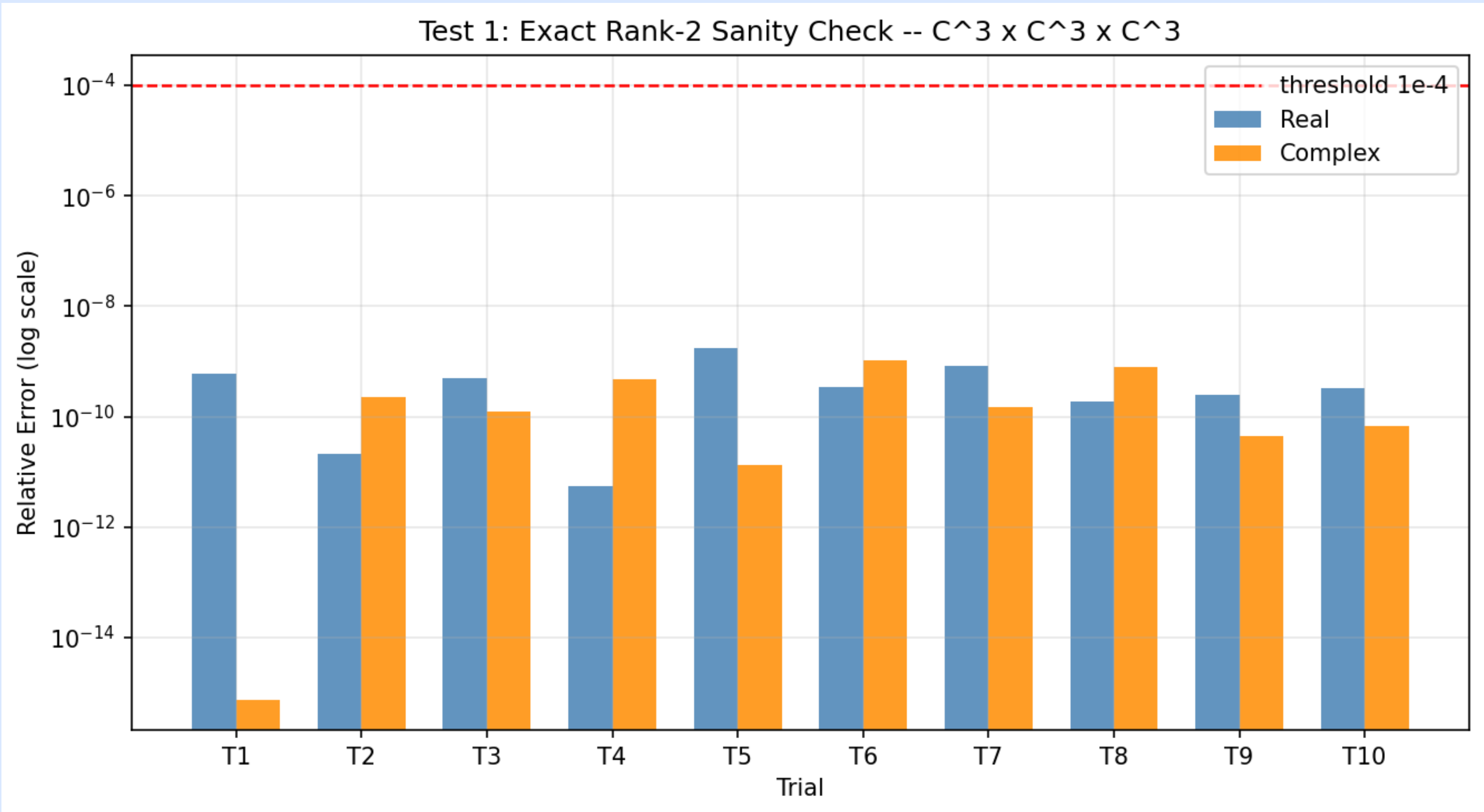
$$\varepsilon_{\text{rel}} = \frac{\|\mathcal{X} - \hat{\mathcal{X}}\|_F}{\|\mathcal{X}\|_F}$$
$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right)$$
$$\text{SSIM} = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}$$

Three validation experiments:

- (i) Exact rank-2 recovery (ii) Noisy rank-2 recovery (iii) Generic random tensor



HOOI converges faster and lower than CP-ALS



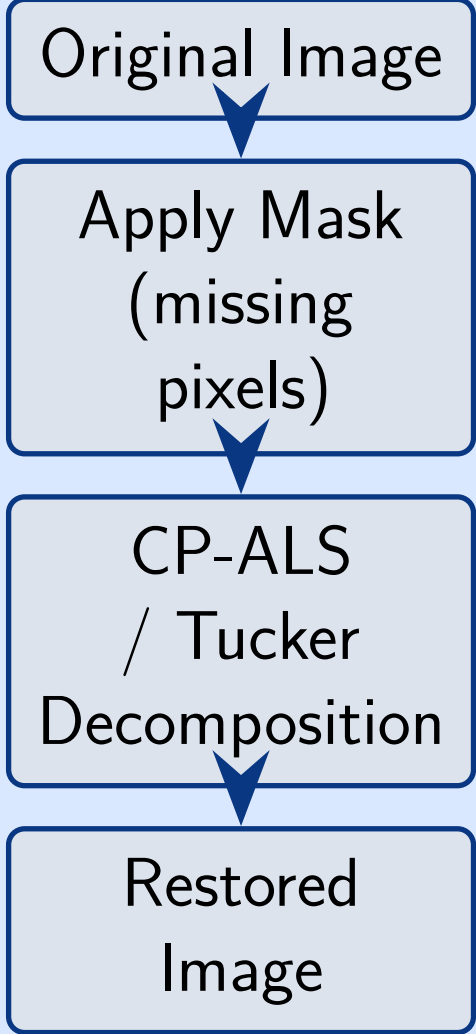
Exact rank-2 recovery: relative error < 10⁻⁴ (all trials)

5. Application: Image Inpainting

RGB image as 3-D tensor:

$$I \in \mathbb{R}^{H \times W \times 3}$$

Inpainting pipeline:



Dataset:

Lena, Baboon, Peppers,
House, Boats, Airplane

figures/reconstruction_panel.png